

STAT452 Practical Midterm

Complete Solution Using R

OLS, Diagnostics, Ridge, LASSO, and Prediction

R-only workflow. Every requested calculation is performed with R. The document shows executable code, the important output, and the conclusion that should be written for each exam part. No calculator or hand-computation route is used.

Contents

1	Reproducible setup	1
1.1	Read and validate the files	1
2	Part (a): Full OLS model and residual analysis	1
2.1	Fit the full model	1
2.2	Model fit and the apparent coefficient paradox	2
2.3	Standard residual plots in R	2
2.4	Normality and constant variance in R	3
2.5	Influence diagnostics in R	3
2.6	Multicollinearity in R	3
3	Part (b): Full model versus noise-only model	4
4	Parts (c)–(f): Shared <code>glmnet</code> setup	4
5	Part (c): Full ridge regression	5
6	Part (d): Ridge predictions	6
7	Part (e): Full LASSO model	6
8	Part (f): Compare ridge and LASSO predictions	7
9	Exam-ready R workflow	8

1 Reproducible setup

The practical dataset contains 38 drivers, eight predictors, and the response `hipcenter`. Chapter 1 of the Notes supplies the OLS and residual-analysis workflow; Chapter 3 supplies cross-validation, ridge, and LASSO.

Run the companion script from the repository root:

```
source("R_training/midterm/practical_test_solution.R")
```

Only `glmnet` is required beyond base R:

```
install.packages("glmnet") # Run once only, if needed
library(glmnet)
```

1.1 Read and validate the files

```
seatpos <- read.csv("R_training/midterm/seatpos.csv")
newdata <- read.csv("R_training/midterm/newdata.csv")

dim(seatpos)           # 38 9
names(seatpos)
str(seatpos)
summary(seatpos)
colSums(is.na(seatpos)) # all zero

stopifnot(nrow(seatpos) == 38, ncol(seatpos) == 9)
stopifnot(identical(
  names(newdata),
  setdiff(names(seatpos), "hipcenter")
))
```

These checks prevent silent errors from a wrong file, altered column name, or missing data. The training data contain no missing values.

2 Part (a): Full OLS model and residual analysis

2.1 Fit the full model

```
full_ols <- lm(hipcenter ~ ., data = seatpos)
summary(full_ols)
confint(full_ols)
```

The coefficient output is:

Term	Estimate	Std. error	p-value
Intercept	436.4321	166.5716	0.014
Age	0.7757	0.5703	0.184
Weight	0.0263	0.3310	0.937
HtShoes	-2.6924	9.7530	0.784
Ht	0.6013	10.1299	0.953
Seated	0.5338	3.7619	0.888
Arm	-1.3281	3.9002	0.736
Thigh	-1.1431	2.6600	0.671
Leg	-6.4390	4.7139	0.182

The fitted R model is

$$\begin{aligned}\widehat{\text{hipcenter}} = & 436.4321 + 0.7757 \text{ Age} + 0.0263 \text{ Weight} \\ & - 2.6924 \text{ HtShoes} + 0.6013 \text{ Ht} + 0.5338 \text{ Seated} \\ & - 1.3281 \text{ Arm} - 1.1431 \text{ Thigh} - 6.4390 \text{ Leg}.\end{aligned}$$

Each slope is interpreted while the other seven predictors are held fixed. For example, the fitted coordinate increases by 0.776 mm per additional year of age, holding weight and all body dimensions fixed. A one-centimeter increase in lower-leg length is associated with a 6.439 mm decrease in the fitted coordinate, holding the other predictors fixed.

The intercept is the fitted coordinate when every predictor equals zero, which is far outside the observed anthropometric range and has no useful physical interpretation.

2.2 Model fit and the apparent coefficient paradox

R reports

$$R^2 = 0.687, \quad R_{\text{adj}}^2 = 0.600, \quad F(8, 29) = 7.94, \quad p = 1.31 \times 10^{-5}.$$

The predictors are highly useful jointly, yet none of the eight individual slopes has $p < 0.05$. This is expected under severe multicollinearity: correlated predictors inflate individual standard errors even when the group explains the response well.

2.3 Standard residual plots in R

```
old_par <- par(mfrow = c(2, 2))
plot(full_ols)
par(old_par)
```

Interpret the four R plots as follows:

- **Residuals vs Fitted:** check linearity and look for a random cloud around zero.
- **Normal Q-Q:** points near the reference line support normality.
- **Scale-Location:** a roughly horizontal band supports constant error variance.
- **Residuals vs Leverage:** identify high-leverage observations with large Cook's distance.

2.4 Normality and constant variance in R

```
shapiro.test(resid(full_ols))

# Base-R Breusch-Pagan LM test
predictor_data <- seatpos[setdiff(names(seatpos), "hipcenter")]
bp_aux <- lm(I(resid(full_ols)^2) ~ ., data = predictor_data)
bp_stat <- nrow(seatpos) * summary(bp_aux)$r.squared
bp_df <- length(coef(full_ols)) - 1
bp_p <- pchisq(bp_stat, df = bp_df, lower.tail = FALSE)
c(BP = bp_stat, df = bp_df, p_value = bp_p)
```

The Shapiro-Wilk result is $W = 0.9715$, $p = 0.434$. R therefore gives no evidence against residual normality at $\alpha = 0.05$. The Breusch-Pagan calculation gives $BP = 14.037$, $df = 8$, $p = 0.0808$, so R does not detect significant heteroscedasticity at 5%, although the scale-location plot should still be checked.

2.5 Influence diagnostics in R

```
influence_table <- data.frame(
  row = seq_len(nrow(seatpos)),
  studentized_residual = rstudent(full_ols),
  leverage = hatvalues(full_ols),
  cooks_distance = cooks.distance(full_ols)
)

influence_table <- influence_table[
  order(influence_table$cooks_distance, decreasing = TRUE),
]
head(influence_table, 5)

4 / nrow(seatpos) # Cook's D guide
2 * length(coef(full_ols)) / nrow(seatpos) # leverage guide
```

Row	Studentized residual	Leverage	Cook's D
31	2.390	0.560	0.695
23	1.938	0.270	0.141
35	-2.323	0.183	0.117
33	1.528	0.248	0.082
17	-0.914	0.401	0.063

Row 31 exceeds the rough Cook's-distance guide $4/n = 0.105$ and leverage guide $2k/n = 0.474$, where $k = 9$ fitted coefficients including the intercept. It should be checked and used in a sensitivity analysis, not automatically deleted.

2.6 Multicollinearity in R

```

predictors <- setdiff(names(seatpos), "hipcenter")

vif <- vapply(predictors, function(variable) {
  others <- setdiff(predictors, variable)
  auxiliary <- lm(
    reformulate(others, response = variable),
    data = seatpos
  )
  1 / (1 - summary(auxiliary)$r.squared)
}, numeric(1))

sort(vif, decreasing = TRUE)
kappa(scale(seatpos[predictors]), exact = TRUE)

```

Variable	VIF	Variable	VIF
Ht	333.138	Arm	4.496
HtShoes	307.429	Weight	3.647
Seated	8.951	Thigh	2.763
Leg	6.694	Age	1.998

The standardized-predictor condition number is 59.77. The extreme VIF values for `Ht` and `HtShoes` confirm severe collinearity, explaining the unstable OLS slopes and motivating ridge/LASSO.

3 Part (b): Full model versus noise-only model

Fit the intercept-only model and ask R for the nested-model partial F -test:

```

null_ols <- lm(hipcenter ~ 1, data = seatpos)
anova(null_ols, full_ols)

```

Model	Res.Df	RSS	Df	F	p
Intercept only	37	131639			
Full model	29	41262	8	7.94	1.31×10^{-5}

At $\alpha = 0.05$, R gives $p < 0.05$, so reject the null hypothesis that all eight slopes equal zero. The full model improves significantly on the noise-only model.

4 Parts (c)–(f): Shared `glmnet` setup

`glmnet` requires a numeric predictor matrix rather than a data frame. The same fold IDs must be used for ridge and LASSO to make the comparison fair.

```
library(glmnet)

X <- model.matrix(hipcenter ~ ., data = seatpos)[, -1, drop = FALSE]
y <- seatpos$hipcenter
X_new <- model.matrix(~ ., data = newdata)[, -1, drop = FALSE]

lambda_grid <- seq(1e-3, 50, by = 1e-3)
set.seed(452)
fold_id <- sample(rep(1:10, length.out = nrow(X)))
```

The explicit `fold_id` makes the answer reproducible. A different random fold allocation can produce a slightly different optimal penalty.

5 Part (c): Full ridge regression

Set $\alpha = 0$. The exam's 50,000-point grid is unusually dense, so the current `glmnet` version may warn near the smallest penalties. The selected penalty occurs safely before that path limit.

```
ridge_cv <- suppressWarnings(cv.glmnet(
  x = X,
  y = y,
  alpha = 0,
  lambda = lambda_grid,
  foldid = fold_id,
  type.measure = "mse",
  standardize = TRUE
))

ridge_cv$lambda.min
ridge_cv$lambda.1se
coef(ridge_cv, s = "lambda.min")
plot(ridge_cv)
abline(v = log(ridge_cv$lambda.min), col = "red", lty = 2)
```

R returns

$$\lambda_{\min} = 36.521, \quad \lambda_{1se} = 50.000.$$

Because λ_{1se} equals the grid's upper boundary, a real analysis using the 1-SE rule should expand the upper end beyond 50.

At $\lambda_{\min}$, R reports:

Term	Ridge	Term	Ridge
Intercept	399.992503	Seated	-1.203317
Age	0.473581	Arm	-1.354624
Weight	-0.106186	Thigh	-1.304223
HtShoes	-0.738541	Leg	-3.109081
Ht	-0.737116		

Ridge keeps every slope nonzero but shrinks the unstable OLS estimates. This matches Chapter 3: the

L_2 penalty trades a small amount of bias for lower variance under multicollinearity. `glmnet` standardizes predictors internally, leaves the intercept unpenalised, and returns coefficients on the original variable scales.

6 Part (d): Ridge predictions

```
ridge_prediction <- drop(predict(
  ridge_cv,
  newx = X_new,
  s = "lambda.min"
))
ridge_prediction
```

New row	Ridge prediction (mm)
1	-139.711
2	-112.954
3	-219.990
4	-149.057
5	-228.157
6	-102.596

7 Part (e): Full LASSO model

Set `alpha = 1` and reuse the same grid and folds:

```
lasso_cv <- cv.glmnet(
  x = X,
  y = y,
  alpha = 1,
  lambda = lambda_grid,
  foldid = fold_id,
  type.measure = "mse",
  standardize = TRUE
)

lasso_cv$lambda.min
lasso_cv$lambda.1se
lasso_coef <- as.matrix(coef(lasso_cv, s = "lambda.min"))
lasso_coef

nonzero <- rownames(lasso_coef)[
  lasso_coef[, 1] != 0 & rownames(lasso_coef) != "(Intercept)"
]
nonzero
```


R returns

$$\lambda_{\min} = 5.748, \quad \lambda_{1se} = 20.288.$$

At $\lambda_{\min}$, the nonzero predictors are

Age, Ht, Leg.

Term	OLS	Ridge	LASSO
Intercept	436.432128	399.992503	397.696648
Age	0.775716	0.473581	0.221804
Weight	0.026313	-0.106186	0
HtShoes	-2.692408	-0.738541	0
Ht	0.601345	-0.737116	-2.200605
Seated	0.533752	-1.203317	0
Arm	-1.328069	-1.354624	0
Thigh	-1.143119	-1.304223	0
Leg	-6.439046	-3.109081	-5.468787

The LASSO solution is sparse: five slopes are exactly zero. With strongly correlated height measurements, LASSO may select one representative and discard another, so the selected set can vary with the fold split. Ridge is usually more stable when all correlated measurements carry signal; LASSO is more useful when a compact predictive model is required.

For an optional post-LASSO comparison in R:

```
post_lasso_ols <- lm(
  reformulate(nonzero, response = "hipcenter"),
  data = seatpos
)
summary(post_lasso_ols)
```

The refitted OLS slopes are approximately 0.581 for Age, -2.325 for Ht, and -6.739 for Leg. These are unpenalised estimates and therefore have larger magnitudes than the LASSO coefficients. The refit is a different estimator; it must not be labelled as the LASSO model.

8 Part (f): Compare ridge and LASSO predictions

```
lasso_prediction <- drop(predict(
  lasso_cv,
  newx = X_new,
  s = "lambda.min"
))

prediction_comparison <- data.frame(
  row = seq_len(nrow(newdata)),
  ridge = ridge_prediction,
  LASSO = lasso_prediction,
  difference = ridge_prediction - lasso_prediction
)
```

```
prediction_comparison
```

Row	Ridge	LASSO	Ridge – LASSO
1	-139.711	-130.368	-9.344
2	-112.954	-112.470	-0.485
3	-219.990	-218.053	-1.937
4	-149.057	-151.148	2.091
5	-228.157	-229.846	1.689
6	-102.596	-100.072	-2.524

The two models give similar predictions for five rows; row 1 differs by about 9.34 mm. The prediction file does not contain observed `hipcenter` values, so R can compare the predictions but cannot determine which model is more accurate on these six cases. That requires observed test responses or repeated cross-validation performance.

9 Exam-ready R workflow

1. Read the files and check dimensions, names, structure, and missing values.
2. Fit `lm(hipcenter ~ ., data = seatpos)`.
3. Report coefficient interpretations, R^2 , adjusted R^2 , and the overall F test.
4. Run the four standard R diagnostic plots and calculate influence and VIF.
5. Compare the null and full models with `anova(null_ols, full_ols)`.
6. Build `x` and `x_new` with `model.matrix`.
7. Fix the seed and fold IDs, then use the same folds for ridge and LASSO.
8. State explicitly whether `lambda.min` or `lambda.1se` is used.
9. Extract nonzero LASSO slopes from `coef()`.
10. Predict with both models and compare the resulting values without claiming a winner when test outcomes are unavailable.